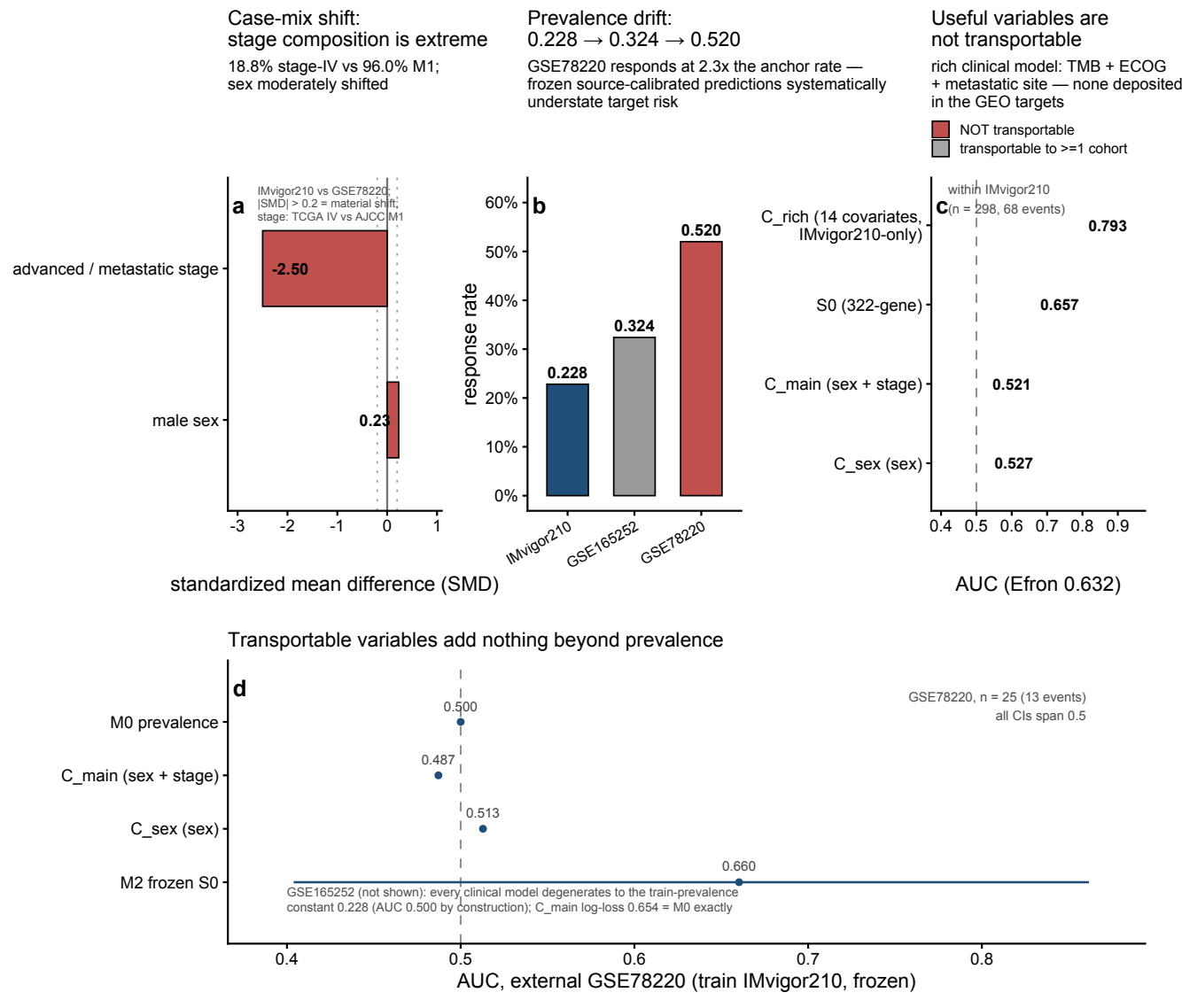
Case-mix shift:
stage composition is extreme
18.8% stage-IV vs 96.0% M1;
sex moderately shifted
Prevalence drift:
0.228 → 0.324 → 0.520
GSE78220 responds at 2.3x the anchor rate — frozen source-calibrated predictions systematically understate target risk
Useful variables are not transportable
rich clinical model: TMB + ECOG + metastatic site — none deposited in the GEO targets
NOT transportable
transportable to >=1 cohort
a
IMvigor210 vs GSE78220;
|SMD| > 0.2 = material shift;
stage: TCGA IV vs AJCC M1
advanced / metastatic stage
male sex
-2.50
0.23
-3
-2
-1
0
1
standardized mean difference (SMD)
b
response rate
60%
50%
40%
30%
20%
10%
0%
0.228
0.324
0.520
IMvigor210
GSE165252
GSE78220
c
within IMvigor210
(n = 298, 68 events)
C_rich (14 covariates, IMvigor210-only)
S0 (322-gene)
C_main (sex + stage)
C_sex (sex)
0.793
0.657
0.521
0.527
0.4
0.5
0.6
0.7
0.8
0.9
AUC (Efron 0.632)
Transportable variables add nothing beyond prevalence
d
GSE78220, n = 25 (13 events)
all CIs span 0.5
M0 prevalence
C_main (sex + stage)
C_sex (sex)
M2 frozen S0
0.500
0.487
0.513
0.660
GSE165252 (not shown): every clinical model degenerates to the train-prevalence constant 0.228 (AUC 0.500 by construction); C_main log-loss 0.654 = M0 exactly
0.4
0.5
0.6
0.7
0.8
AUC, external GSE78220 (train IMvigor210, frozen)